

Relativistic Compactness Parameter ξ Across Astrophysical Regimes

Relativistic correction to Ra_c : $\Delta Ra/Ra \approx \xi$

1e-18%

1e-13%

1e-08%

1e-03%

0.1%

1.0%

10%

3%

$\mathcal{O}(1)$ correction ($\xi = 1/3$)

1% correction ($\xi = 0.01$)

Early universe
(radiation era)

QGP
($T \sim 200$ MeV)

Accretion disk
(inner)

NS core

NS crust

White dwarf

Liquid metals

Laboratory water

Classical regime
Relativistic regime

$\log_{10} \xi$ where $\xi = p_0/(\epsilon_0 c^2)$

-20

-15

-10

-5

0